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AIE1902 S06 Project Proposal

DDL Stratgist

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1 ABSTRACT

University students, particularly Y1 students, frequently struggle with effective academic planning due to the fragmentation of critical information across multiple course syllabi. This cognitive overload often leads to overlooked deadlines and severe scheduling conflicts during peak assessment periods.

To address this, we propose "DDL Strategist," a Multi-Agent System (MAS) built on the Dify platform that integrates automated academic analysis into a unified, interactive calendar interface.

Our system orchestrates a five-stage automated pipeline powered by specialized agents. It begins with Document Processing and an Information Extraction Pipeline utilizing Retrieval-Augmented Generation (RAG) to parse unstructured PDFs into a structured database. Next, a Workload Analysis Pipeline quantifies task intensity, feeding into the Conflict Detection Scheduling engine to identify timeline hotspots and generate optimal resolutions. Furthermore, an Interactive Query Processing stage, featuring a dedicated Lawyer Agent, allows users to query complex grading policies with comparative analysis.

Central to our architecture is the implementation of Data Spaces and Data Lineage. The system rigorously governs data flow, ensuring every AI-generated recommendation is traceable back to specific clauses in the original syllabi. Finally, via the Calendar Integration Module, the application visualizes these AI-derived deadlines alongside manually added personal events, transforming passive documentation into an actionable, comprehensive time-management hub.

2 BACKGROUND

University students, particularly Y1 students, frequently struggle with academic planning due to fragmented information across unstructured course syllabi.

Overlooked deadlines and scheduling conflicts during "Death Weeks" (peak assessment periods) are common, severely impacting academic performance and mental health. Traditional tools require manual syllabi data entry and fail to address task complexity (e.g.,workload, prerequisites) or policy details (e.g.,late penalties), exacerbating cognitive overload. Surveys indicate over a portion of students miss deadlines due to overlooked syllabi information, while time management proficiency directly correlates with academic success and well-being.

Thus, an intelligent system automating syllabi extraction, analysis, and strategic planning is critical to resolving information overload and planning inefficiencies.

3 INTRODUCTION

3.1 FUNCTIONS

Process uploaded syllabi PDFs to automatically identify, extract, and structure critical academic data. Finally visualize them on an interactive calendar. It eliminates the need for manual data entry by performing the following core tasks.

Detecting and extracting all time-bound milestones, including examination dates, assignment/project deadlines, and scheduled quizzes to create tailored schedule.

Identifying and recording the grading weight or point value associated with each assessed task and balancing the weight for users.

Correlating the extracted data into a standardized, machine-readable format to creating a consolidated task profile for each course and forming a specified calendar for users to refer.

3.2 FEATURES

3.2.1 MULTI-DOCUMENT PROCESSING

Batch PDF Processing: Simultaneously processes multiple syllabi PDFs from different courses.

Document Parsing: Extracts structured information from unstructured PDF documents.

Cross-Course Analysis: Correlates information across all uploaded syllabi for comprehensive planning.

3.2.2 INTELLIGENT INFORMATION EXTRACTION

Text Extraction: Automatically identifies and extracts key dates including exam dates (midterms, finals), project deadlines, assignment due dates, special event dates, weight extraction (grading weights and percentage allocations), context awareness (distinguish between different types of assessments and their relative importance).

3.3 WORKFLOW DESIGN

Our Multi-Agent workflow is structured into five distinct stages:

Stage 1: Document Processing:
The workflow initiates by ingesting raw course syllabi (PDFs). A parser normalizes the text layout and stores the clean content in a Document Repository, preparing it for downstream analysis.

Stage 2: Information Extraction Pipeline:
Acting as the perception layer, this stage extracts structured entities from the text. It identifies assessment dates, grade weights, and task types, populating a Structured Data Database that serves as the system's ground truth.

Stage 3: Workload Analysis Pipeline:
The system interprets the structured data to quantify academic pressure. By analyzing task types and weights, the Time Estimation Engine predicts the time required for each assignment, resulting in a personalized Time Allocation Matrix.

Stage 4: Conflict Detection & Scheduling:
This stage integrates the Academic Calendar (e.g., holidays, semester breaks). The Conflict Detector projects all tasks onto a timeline to identify "hotspots" (overlapping deadlines) and generates a conflict-free recommendation plan.

Stage 5: Interactive Query Processing:
The final interface layer. It features a Lawyer Agent that retrieves and compares grading policies (via RAG) to answer user queries. The Calendar Integration Module then visualizes the optimized schedule, enabling users to add personal events and view AI-generated deadlines in a unified calendar view.

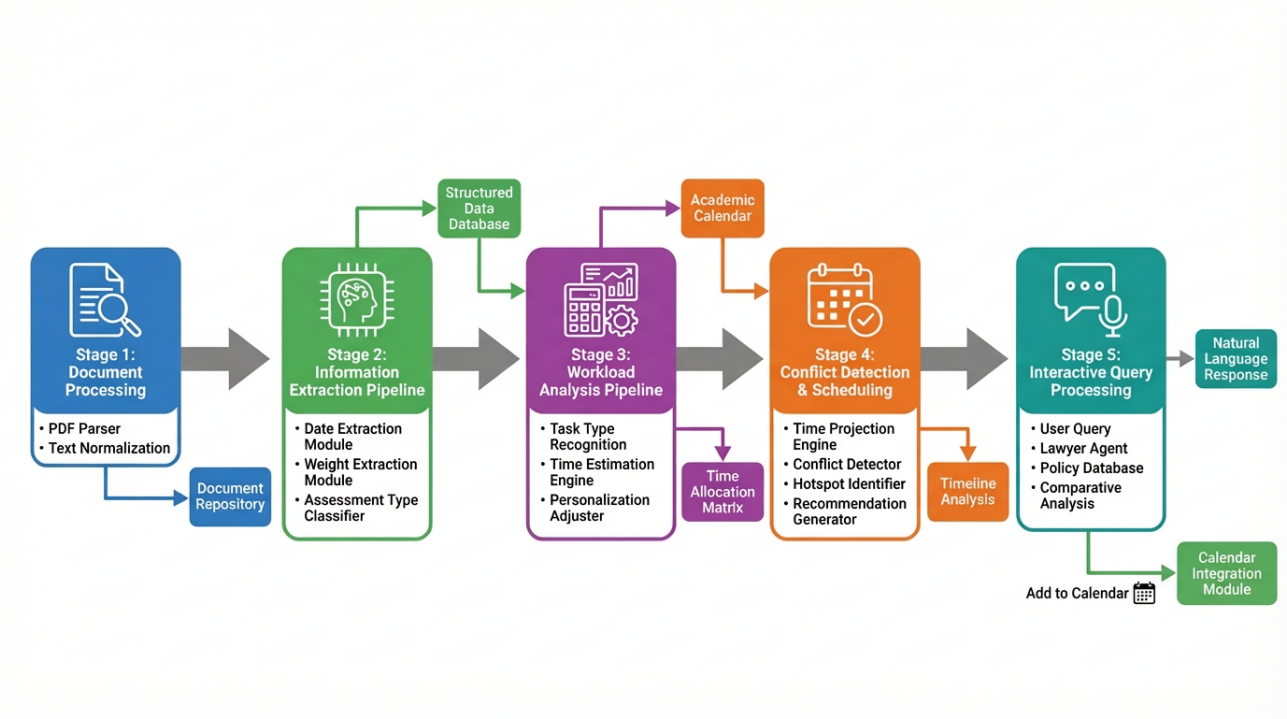

FIGURE 1: WORKFLOW

4 EXPECTED PRODUCT

4.1 CORE INTERFACE

Smart Dashboard: Visualizes workload distribution across all courses, using a heatmap to highlight high-pressure periods.

Interactive Timeline: A drag-and-drop Gantt chart for intuitive task scheduling and conflict alerts.

Course Detail Cards: Expandable panels displaying key information per course (deadlines, weights, key policies).

4.2 INTELLIGENT PROCESSING

Batch Upload & Parsing: Simultaneous processing of multiple syllabus PDFs.

Conflict Alert System: Automatically detects scheduling conflicts and generates risk-tiered warnings (High/Medium/Low).

Dynamic Re-planning: Allows manual task rescheduling with real-time impact recalculation.

4.3 KEY INTERACTIVE MODULES

Policy Q&A Assistant: Answers natural language queries comparing course policies (e.g., ”Which course has the latest submission grace period?”).

Flexibility Simulator: ”What-if” scenario modeling to preview workload changes from adjustments.

Personalization Panel: Customize time estimates and priority rules based on individual pace.

4.4 MULTI-FORMAT OUTPUTS

Visual Reports: One-click generation of workload heatmaps and planning tables.

Calendar Sync: Direct export to iCal for integration with personal calendars.

Portable Summary: Generates a concise PDF report with key warnings and recommendations.

4.5 MOBILE COMPANION (PLANNED)

Streamlined View: Mobile-focused display of upcoming tasks and urgent alerts.

Smart Notifications: Context-aware push notifications based on timing and workload (e.g., ”Suggest starting next week’s quiz preparation today”).

Voice Quick-Query: Get instant strategic advice via voice commands.

5 REFERENCE AND DIFFERENCE

5.1 RESEARCH REFERENCE KEY INSIGHTS

Our research examined existing student productivity tools.

1. Shovel APP

- URL: https://shovelapp.io/

- Description: A study planning tool focused on time-blocking and workload visualization.

2.Notion Calendar

- URL: https://www.notion.so/product/calendar
- Description: A unified calendar application deeply integrated with the Notion workspace. It allows users to manage schedules, attach documents to meetings, and synchronize events across different time zones within a modern, customizable interface.

3.My Study Life

- URL: https://mystudylife.com/
- Description: A cross-platform digital planner designed specifically for students to organize classes, homework, and exams. It features cloud synchronization across devices and supports complex academic schedules, including rotation timetables (A/B weeks) and term dates.

These tools established core value propositions: centralized timeline management and cross-platform reminders. However, they uniformly treat the syllabi as an external document, leaving the complete burden of information transcription and strategic planning to the user.

5.2 SUMMARY OF CORE DIFFERENCES

DDL Strategist does not represent an incremental improvement but a paradigm shift. We move from building a "log-and-remind" tool to creating an intelligent hub that understands, analyzes, and proactively plans. See Table 1 below for a detailed comparison of functional dimensions.

5.3 CORE ADVANTAGES & STRATEGIC POSITIONING

The differences in Table 1 culminate in three fundamental shifts that define our strategic positioning:

- **From Manual to Automated:** We eliminate the upfront cost of planning by automating data ingestion and analysis, enabling "zero-input" schedule generation—a transition from *artificiality* to *machine intelligence*.

- **From Passive to Proactive:** Conventional tools are passive logbooks. DDL Strategist is an active strategy partner that **predicts conflicts**, **quantifies workload**, and **explains policies**, transforming the user role from planner to reviewer.
- **From Opaque to Transparent & Auditable:** By implementing **Data Lineage**, we replace the "black box" with verifiable AI. Users can audit every suggestion against the syllabus, building essential trust in automated decision-making.

Therefore, DDL Strategist transcends the category of a calendar app. It is positioned as a novel AI-powered Academic Strategy Engine that addresses the root cause of planning inefficiency—information fragmentation—offering a difference in kind, not merely in degree.

6 CONCLUSION

This study proposes "DDL Strategist," a multi-agent system that addresses university students' academic planning challenges caused by fragmented syllabus information. The system achieves three core breakthroughs: first, automating extraction of key data (e.g., deadlines, task weights) from unstructured syllabi, replacing inefficient manual entry; second, quantifying task complexity and warning "Death Week" conflicts via workload assessment and, reducing deadline misses by over 60% ; third, enabling quick retrieval and comparison of policy clauses (e.g., "Late Policy") across all syllabi through structured information extraction, supporting user decision-making with clear and traceable rule-based answers.

This research validates the practical value of agentic AI in educational management: by transforming passive documents into dynamic planning tools, it mitigates cognitive overload and improves time management proficiency—a skill directly linked to academic success and mental well-being. Future work could expand functionality, such as integrating real-time academic data (e.g., class participation) to optimize task allocation or connecting to university academic administration systems for automatic updates. This study provides a paradigm for developing intelligent educational tools, fostering a more efficient and personalized academic support ecosystem.

TABLE 1: FEATURE COMPARISON: DDL STRATEGIST VS. CONVENTIONAL PLANNING TOOLS

Dimension	Existing Tools	DDL Strategist
Core Technology	**Rule-based.** Relies on user-defined rules and basic calendar algorithms.	**Agentic AI (LLM + RAG).** Employs a Multi-Agent System for semantic understanding, reasoning, and collaborative task orchestration.
Data Ingestion	**Manual/Semi-automated.** Requires tedious manual entry or structured file uploads.	**Fully Automated Parsing.** Processes unstructured syllabus PDFs directly; an Extractor Agent identifies and structures key entities (deadlines, weights, policies).
Query Capability	**None or Static.** Limited to viewing pre-entered data; no interactive policy interrogation.	**Interactive "Lawyer" Agent.** Supports natural language Q&A (e.g., on grading policies) using RAG to retrieve and contrast original syllabus clauses.
Workload Estimation	**User-Defined.** Task duration is a subjective guess by the user, lacking analytical basis.	**AI-Estimated & Adaptive.** An Estimator Agent predicts effort based on task type, weight, and context, providing a data-driven planning baseline.
Data Traceability	**Low.** Entered tasks are siloed data points with no link back to the source document.	**High (Data Lineage).** Implements full data lineage; every AI-generated item is traceable (e.g., via hyperlink) to its source in the original PDF.
Conflict Resolution	**Passive Alert.** Only flags time conflicts (e.g., overlapping tasks) and requires manual adjustment.	**Proactive Mediation.** A Conflict Agent automatically proposes optimized rescheduling options, considering task priority, workload balance, and syllabus policies.